

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2024

Subject: 031001121 Chemistry I

Section: 15-18

Date: 3 October 2024

Time: 13.00 – 15.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 parts. There will be 9 pages (including this page) and they are worth 70 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are NOT allowed in the examination.
- Periodic table is attached on the last page of the midterm examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Exam
CLO 1	Explain principle of chemistry, chemical change and physical change of matter and phenomena.	Questions in Parts 1-2.
CLO 2	Explain classification of matter and selection of proper separation method. Explain energy change of system.	Questions in Parts 1-2.
CLO 3	Explain atomic structure and electron configuration to explain any scientific phenomena.	Questions in Parts 1-2.
CLO 4	Explain and calculate relationship of any matters in chemical reaction to apply in workplace and in scientific industry.	Questions in Parts 1-2.
CLO 5	Apply basic chemistry to other subjects related.	Questions in Parts 1-2.
CLO 6	Realize to have self-discipline, be on time, and be responsible to yourself and society.	Questions in Parts 1-2.

Instructors: Assoc. Prof. Dr. Nathjanan Jongkon and Asst. Prof. Dr. Sunanta Chuayprakong

	Part 1	Part 2	Total
score			

Part 1. Multiple Choice (10 points)

Please mark your answer with (x) in the table provided for the multiple choice questions.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

- What is the definition of "close system"?
 - energy is not exchanged.
 - energy and mass are not exchanged.
 - mass is not exchanged.
 - an example of this system is a jar.
- Which statement is incorrect?
 - A boiled soup is an example of a system where everything can exchange with its environment.
 - A system where energy can leave the system is called a closed system.
 - A system where only quantity can exchange with its environment is called an isolated system.
 - A cooler is an example of a system where nothing can enter or leave the system.
- What is the percentage composition of carbon in CH_3COOH ?
 - 6.67 %
 - 20.00 %
 - 40.00 %
 - 53.33 %
- In the compound ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$, which one is incorrect?
 - Molar mass is 149 g/mol
 - % N is 28.20%
 - % P is 20.80%
 - Mass of nitrogen/mol of compound = 28 g
- What is the volume, in liters, of 0.500 mol of C_3H_8 gas at STP?
 - 1.12 L
 - 2.24 L
 - 11.2 L
 - 22.4 L
- Which of the following samples contains the largest number of atoms?
 - 24.0 grams of carbon
 - 12.04×10^{23} N_2 molecules
 - 1.50 mol NH_3
 - 28.00 grams of CO
- Which compound has the highest molar mass?
 - $(\text{NH}_4)_3\text{PO}_3$
 - Sr_3N_2
 - CsOH
 - NaClO_3
- $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$
 How many moles of carbon dioxide are produced from the combustion of 80 g of CH_4 ?
 - 2.50 mol
 - 5.00 mol
 - 7.50 mol
 - 14.00 mol
- What is the summation of stoichiometric coefficients for the following chemical reaction?

$$\text{Mg}(\text{OH})_2(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$$
 - 0
 - 4
 - 5
 - 6
- $\text{N}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$
 How many moles of hydrogen are needed to react with 2 moles of nitrogen?
 - 1 mole
 - 2 moles
 - 3 moles
 - 6 moles

Part 2 Write your answer in the space provided for each question. Calculation detail must be provided for questions in which CALCULATION is noted.

1. Please read the following sentences and indicate whether they are true or false by writing True or False in the space provided, respectively. (5 points)

..... 1. A reactant breaks down to form simpler compounds in a combination reaction.

..... 2. There is no difference in mole of 1 g of hydrogen and 1 g of hydrogen gas.

..... 3. NH_3 0.5 mol weighs 8.5 g.

..... 4. Water and carbon monoxide are always products of a complete combustion reaction.

..... 5. Bromine has two isotopes, Br-79 and Br-81. Relative atomic mass of bromine is 80, if both exist in equal amounts.

2. What is the definition of an open system and a closed system and what are the differences between them? (2 points)

3. Please classify and write the following as open, closed, or isolated system. (5 points)

..... a. A cold beverage kept in a closed thermos flask.

..... b. A chemical reaction taking place in an enclosed flask.

..... c. A piece of paper is burnt.

..... d. Boil water in a beaker.

..... e. Helium gas filled in a pink balloon..

4. Please calculate the percentage and the relative atomic mass in the following questions below.

4.1 Lithium has two naturally occurring isotopes: Li-6 amu (10% abundance) and Li-7 amu (90% abundance). Calculate its relative atomic mass.

(CALCULATION, 3 points)

- 4.2 Copper has two isotopes (^{63}Cu and ^{65}Cu). The relative atomic mass of copper is 63.50. Calculate the percentage abundance of the two isotopes.
(CALCULATION, 3 points)

5. Determine the molecular weight and %hydrogen for the following molecules.
(CALCULATION, 6 points)

a. LiH

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b. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

c. Glycine ($\text{H}_2\text{NCH}_2\text{COOH}$)

6. Calculate the number of **moles** of each compound. (CALCULATION, 6 points)

a. 2.24 L of H_2 gas at STP

b. 1.806×10^{26} NO_2 molecules

c. CO 140 mg

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7. Complete the table below. (CALCULATION, 6 points)

	CO_2 11 g
Amount in mole	
Number of O atoms	
Amount in grams of C	
Volume at STP	

8. Write a balance chemical equation for the following reactions. (4 points)

a. The combustion of ethane (C_2H_6)

b. Ammonia reacts with oxygen gas to produce nitrogen monoxide and water

9. Balance the following reactions and specify whether they are synthesis, decomposition, neutralization or combustion reaction. (8 points)

Reaction	Reaction type
..... $\text{NaN}_3 (\text{s}) \rightarrow$ $\text{Na} (\text{s}) +$ $\text{N}_2 (\text{g})$	
..... $\text{C}_4\text{H}_8 (\text{g}) +$ $\text{O}_2 (\text{g}) \rightarrow$ $\text{CO}_2 (\text{g}) +$ $\text{H}_2\text{O} (\text{g})$	
..... $\text{Mg} (\text{s}) +$ $\text{O}_2 (\text{g}) \rightarrow$ $\text{MgO} (\text{s})$	
..... $\text{LiOH} (\text{aq}) +$ $\text{H}_2\text{SO}_4 (\text{aq}) \rightarrow$ $\text{Li}_2\text{SO}_4 (\text{s}) +$ $\text{H}_2\text{O} (\text{l})$	

10. Balance the equation by writing the numeric ratio in the space provided.

- a. ____ $\text{N}_2 (\text{g})$ ____ + ____ $\text{H}_2 (\text{g}) \rightarrow$ ____ $\text{NH}_3 (\text{g})$ (1 point)
- b. What volume of NH_3 will be produced if you use 4.48 L of N_2 ? Assume the reaction is performed at STP. (CALCULATION, 2 points)

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- c. What mass of NH_3 will be produced if you have 1 g of H_2 ?
(CALCULATION, 2 points)

- d. If 4.816×10^{23} particles of NH_3 were produced how many moles of N_2 reacted?
(CALCULATION, 2 points)

11. The rusting of iron results in the production of iron (III) oxide: $\text{Fe}_{(s)} + \text{O}_{2(g)} \rightarrow \text{Fe}_2\text{O}_{3(s)}$
- Determine the volume of oxygen gas at STP conditions which would be needed to completely oxidize 168 g of iron and
 - Calculate the mass of rust formed.
- (Hint: use atomic weight of O = 16 and Fe = 56 g/mol) (CALCULATION, 5 points)

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Periodic Table of the Elements

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1 hydrogen 1 H 1.0079																	18 helium 2 He 4.0026	
2 lithium 3 Li 6.941	4 beryllium 4 Be 9.0122											13 boron 5 B 10.811	14 carbon 6 C 12.011	15 nitrogen 7 N 14.007	16 oxygen 8 O 15.999	17 fluorine 9 F 18.998	10 neon 10 Ne 20.180	
11 sodium 11 Na 22.990	12 magnesium 12 Mg 24.305											13 aluminum 13 Al 26.982	14 silicon 14 Si 28.086	15 phosphorus 15 P 30.974	16 sulfur 16 S 32.065	17 chlorine 17 Cl 35.453	18 argon 18 Ar 39.948	
19 potassium 19 K 39.098	20 calcium 20 Ca 40.078	3 scandium 21 Sc 44.956	4 titanium 22 Ti 47.867	5 vanadium 23 V 50.942	6 chromium 24 Cr 51.996	7 manganese 25 Mn 54.938	8 iron 26 Fe 55.845	9 cobalt 27 Co 58.933	10 nickel 28 Ni 58.693	11 copper 29 Cu 63.546	12 zinc 30 Zn 65.39	13 gallium 31 Ga 69.723	14 germanium 32 Ge 72.61	15 arsenic 33 As 74.922	16 selenium 34 Se 78.96	17 bromine 35 Br 79.904	18 krypton 36 Kr 83.80	
37 rubidium 37 Rb 85.468	38 strontium 38 Sr 87.62	39 yttrium 39 Y 88.906	40 zirconium 40 Zr 91.224	41 niobium 41 Nb 92.906	42 molybdenum 42 Mo 95.94	43 technetium 43 Tc [98]	44 ruthenium 44 Ru 101.07	45 rhodium 45 Rh 102.91	46 palladium 46 Pd 106.42	47 silver 47 Ag 107.87	48 cadmium 48 Cd 112.41	49 indium 49 In 114.82	50 tin 50 Sn 118.71	51 antimony 51 Sb 121.76	52 tellurium 52 Te 127.60	53 iodine 53 I 126.90	54 xenon 54 Xe 131.29	
55 caesium 55 Cs 132.91	56 barium 56 Ba 137.33	57-70 ■	71 lutetium 71 Lu 174.97	72 hafnium 72 Hf 178.49	73 tantalum 73 Ta 180.95	74 tungsten 74 W 183.84	75 rhenium 75 Re 186.21	76 osmium 76 Os 190.23	77 iridium 77 Ir 192.22	78 platinum 78 Pt 195.08	79 gold 79 Au 196.97	80 mercury 80 Hg 200.59	81 thallium 81 Tl 204.38	82 lead 82 Pb 207.2	83 bismuth 83 Bi 208.98	84 polonium 84 Po [209]	85 astatine 85 At [210]	86 radon 86 Rn [222]
87 francium 87 Fr [223]	88 radium 88 Ra [226]	89-102 ■	103 lawrencium 103 Lr [262]	104 rutherfordium 104 Rf [261]	105 dubnium 105 Db [262]	106 seaborgium 106 Sg [266]	107 bohrium 107 Bh [264]	108 hassium 108 Hs [269]	109 meitnerium 109 Mt [268]	110 darmstadtium 110 Ds [271]	111 roentgenium 111 Rg [272]	112 ununbium 112 Uub [277]	113 ununtrium 113 Uut [284]	114 ununquadium 114 Uuq [289]	115 ununpentium 115 Uup [288]	116 ununhexium 116 Uuh [292]	117 ununseptium 117 Uus [294]	118 ununoctium 118 Uuo [294]

■ Lanthanide series

■ Actinide series

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

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